CONSTRUCTIVE QUESTION-LEVEL EXCHANGE

A sharp tail boundary for recoverable information

Finite seeds, local correction, and a delivered answer

5 November 2026

Selected completed target. For instantaneous Poisson cube bursts in d dimensions with $\mathbb{P}(R > r) = r^{-\alpha}$, determine which tail exponents permit a fixed finite input-dependent seed to retain a classical bit for every requested finite horizon, using bounded local state, speed and density, a uniform finite circuit, and a paid local readout. Immutable gate laws and an input-independent schedule are granted as in P3's final refinement.

Result. At sufficiently small positive intensity, the operation is possible precisely when $\alpha > d + 1$. For $\alpha \le d + 1$, the accepted complete-cone erasure theorem forbids it at every positive intensity. The construction here supplies the positive direction, with output error at most $1/4$ uniformly in the requested horizon.

Quantitative operation. For a family-dependent exponent $1 < z < \alpha - d$, the mutable line uses $O(T^{1/z})$ cells, delivery takes $O(T)$ time, and local work is $O(T^{1+1/z})$. The exponent approaches the necessary extent exponent $1/(\alpha - d)$ from above. Constants and the admitted intensity can be severe.

P2 response. Finite noisy position measurements and additive force synthesize a momentum replacement independently of a hidden constant force and old reference records in nominal time $1.2/\gamma$, without learning the force. A more expensive route directly prepares the original sine-biased arms in $1.1/\gamma$. Uniform interval certificates cover the entire soft-spring range. Preparation, calibration, cooling, switching and reuse remain material costs.

P1 retained and strengthened. The accepted thermal-loading compiler remains at scope. An extensive Fano-minor certificate for the actual BDL check matrices excludes both logarithmic graphic/cographic defect guarantees and the regular-matroid sampler route. It does not prove slow mixing or sampling hardness.

Status. This is a mathematical construction with reproducible finite certificates, not a physically built device or an established historical-priority claim. Complete proof notes, independent audits and code accompany this report. The selected theorem concerns strict Pareto tails and the granted circuit primitives; it does not claim autonomous control or the general finite-$(d + 1)$-moment endpoint.

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1 The consequence and the choice

P1 had already supplied a conditional efficient write/read route for the proved random-embedding passive quantum-memory family. P2 had a campaign comparison but not its optimal physical price. P3 had an architecture-independent impossibility theorem but no composable positive information operation.

	Earned input	New result	Consequence
P1	Syndrome-history sampler and loading identity	Extensive Fano packing in physical matrices	Two sampler guarantees fail structurally
P2	Entropy composition and amortized waiting	Finite instrument and causal force reset	Inference time is not an intervention lower bound
P3	Finite-seed cone-loss theorem	Local burst correction, growth and readout	Sharp strict-Pareto tail boundary

I select the strict-Pareto question because its positive construction closes a foundational classification. It determines when local correction can overcome unbounded spatial disasters and when every finite-speed architecture must lose its original information.

The prior equilibrium-inference, passive quantum-memory and partial-SWAP results remain at their accepted scopes. Their originals are unchanged. The partial-SWAP compressor retains its weak-angle memory and synthesis costs; the unrestricted joint memory/processing law remains unresolved. None of these statements is upgraded to general physical robustness here.

2 The exact theorem

Fix $d \geq 1$. Centers and times form a Poisson process of intensity $\lambda\, dz\, dt$. Independent radii obey $\mathbb{P}(R > r) = r^{-\alpha}$ for $r \geq 1$, with $\alpha > d$. A burst instantaneously corrupts all mutable registers in $z + [-R, R]^d$, with arbitrary input-dependent values. The finite dth moment makes the local hit rate finite.

Hardware has bounded density, degree, local state, arity, range, rate and information speed. A fixed Boolean basis supplies COPY, reset and local computation. Workspace, messages, outputs and mutable control records remain corruptible. Immutable wiring, laws and an input-independent schedule may depend on T, as granted by P3's final refinement; one uniform algorithm produces them, and description/preparation costs count. Their reliability is a premise.[2]

Every input-dependent variable starts inside a fixed finite seed of radius r_0, independent of T. Horizon-dependent input-independent hardware contains no copy of the bit. A finite local decoder delivers it at $S(T) \geq T$, and every fault exposure through delivery counts.

Theorem 1 (Sharp tail exponent at low intensity)**.** *If $d < \alpha \leq d + 1$, every positive intensity prevents a horizon-uniform output error at most $1/4$ from a fixed finite input seed. If $\alpha > d + 1$, a uniform finite circuit family achieves that operation for all sufficiently small positive intensities.*

For fixed integers B, U satisfying

$$U \geq c_0 B, \qquad z = \log U / \log B \in (1, \alpha - d),$$

the active binary line has $O(T^{1/z} + 1)$ cells, total delivery takes $O(T + 1)$ time, and local work is $O(T^{1+1/z} + 1)$. Constants and the admitted intensity depend on the chosen family.

This is a low-intensity tail classification, not an exact critical-intensity calculation or a theorem for all non-Pareto laws with finite $\mathbb{E}R^{d+1}$.

3 Necessity

The accepted P3 coupling confines the two input worlds' difference to a cone of radius $r_0 + vt$. A burst covering it can erase all input-dependent registers to common values. With

$$\Lambda(S) = \lambda 2^d \int_0^S \mathbb{E}(R - r_0 - vt)_+^d \, dt,$$

the complete outputs obey

$$\mathrm{TV}(P_0^S, P_1^S) \le e^{-\Lambda(S)}, \quad P_{\mathrm{error}} \ge \tfrac{1}{2}(1 - e^{-\Lambda(S)}).$$

This holds uniformly over the permitted horizon-specific schedules.[2] For $a \ge 1$, Pareto integration gives

$$\mathbb{E}(R - a)_+^d = \frac{d!}{\prod_{j=1}^d (\alpha - j)} a^{d-\alpha}.$$

Thus $\Lambda(S)$ diverges exactly for $\alpha \le d + 1$. At equality, for $r_0 \ge 1$,

$$\Lambda(S) = \frac{\lambda 2^d}{v} \log \frac{r_0 + vS}{r_0}.$$

A finite error rate at each cell therefore does not suffice. A fixed-extent device also has ceiling $T \lesssim L^{\alpha-d}/\lambda$ at fixed accuracy. Our construction approaches this extent exponent; it does not prove a universal cell-count optimum for sparse or moving architectures.

4 The central operation and the existing comparator

Gacs supplies separated spatial replicas and repeated temporal computation.[3] The closer comparator is the hardwired Tsirelson/exRec machinery of Balasubramanian–Davydova–Lake, which handles arbitrary bad-gadget outputs and incoming damage, and explicitly discusses repeated correction.[4] These are credited existing principles.

Its p-bounded theorem does not directly apply to Pareto bursts: one event corrupts many adjacent cells with polynomial probability, violating a bound p^m for a fault set of size m. The fixed Tsirelson space/time factors $3, 6$ also do not automatically approach the causal exponent. The needed bridge is a bounded-island circuit with adjustable block length B and time $O(B)$.

4.1 Two rounds, six passes

Five separated Info bits encode one bit. A regular word has at least four agreeing copies; workspace may be arbitrary. A nice fault set fits within one spatial interval and one temporal interval of fixed width w.

A three-pass round freezes Info. Each pass resets Work, copies neighboring input replicas into fixed message lanes, routes locally, majority-decodes, computes the desired Boolean function, and broadcasts to five Hold positions. Three Hold tracks and gaps longer than w separate the passes. Each output Info takes its local three-Hold majority. All spatial colors finish a pass before the next starts; all outputs commit only after the layer is ready.

With regular inputs, one old wrong replica and one new spatial corruption still leave three correct copies. One island spoils at most one entire pass and corrupts all retained Holds at at most one output position. Other positions receive two correct Holds. Output is regular and logically correct.

Prepend one normalizing three-pass round. If the island hits normalization, the subsequent clean logical round produces a perfect repetition from the resulting fixed words. If it hits the logical round, normalization supplied perfect inputs. Gap/commit faults affect one replica. We obtain:

R. Nice macrosteps regularize arbitrary incoming registers.
C. Nice macrosteps correctly update regular input encodings.

Let $I_t(i)$ mark irregular Info and $\mathrm{Bad}(i,t)$ a non-nice macrostep window. Then

$$I_{t+1}(i) \le 1_{\mathrm{Bad}(i,t)}, \qquad E' \subseteq \mathcal{P}(E) := \mathrm{Bad}(E) \cup \mathcal{N}^+ \mathrm{Bad}(E).$$

Here $\mathcal{N}^+$ advances one step and expands by the fixed input neighborhood. This is a monotone, value-independent support bound for every raw fault field. A damaged encoding cannot leave an indefinitely propagating invalid structure.

Fixed lanes, coloring and nearest-neighbor COPY/SWAP give $O(B)$ time. Flatten tracks into adjacent binary sites; Info moves only within a fixed microcell, while long routes carry copies. Recursively compile every binary site, including workspace. The alphabet stays fixed. Padding uses explicit identity/reset gates, themselves protected at the next level. Hence $L_k = B^k$ and $\tau_k = U^k$.

Choose $U = \lceil B^{z_*} \rceil$ with $1 < z_* < \alpha - d$ and sufficiently large B. The exact exponent $z = \log U / \log B$ meets the strict inequalities. Full routing and boundary details are in the formal compiler supplement.

5 Probability under the actual burst law

Restrict the circuit to a line. Integrating transverse burst centers gives the marked intensity

$$\lambda (2R)^{d-1}\, dx\, dt\, F(dR).$$

The line sees a radius law tilted by R^{d-1}, with effective exponent $\alpha - d + 1$. Using the original one-dimensional law would give the wrong threshold.

Define $E_{k+1} = \mathcal{P}(E_k)$. A specified level-$k$ test depends on raw faults in a local spacetime box. Including the preceding-period and neighbor guard, its half-spans satisfy

$$\rho_k \le aL_k + \rho_{k-1}, \qquad \eta_k \le b\tau_k + \eta_{k-1}$$

for fixed a, b. Their ratios to L_k, τ_k are uniformly bounded by geometric sums. Choose w larger than twice those bounds and a radius guard, then choose B larger again. A non-nice parent contains two child tests sufficiently separated in space or time.

Let p_k bound any specified level-k support-test probability. First charge every burst with $R \ge cL_k$ intersecting an enlarged parent region. Its mean count is at most

$$\lambda \tau_{k+1} \mathbb{E}[(AL_{k+1} + 2R)(2R)^{d-1} 1_{\{R \ge cL_k\}}] \le D\lambda\theta^{k+1}, \quad \theta = UB^{d-\alpha}.$$

After removing those marks, two separated child tests have disjoint primitive Poisson influence regions. Spatial separation uses the radius cutoff; temporal separation uses instantaneous bursts. The small-radius fields are independent. Monotonicity bounds each test probability by p_k, without conditioning on successful data history.

There are at most $C(B,U)$ candidate pairs, hence

$$p_{k+1} \le Cp_k^2 + D\lambda\theta^{k+1}.$$

The square follows from disjoint primitive regions, not multiplied marginal estimates. The base bound is $p_0 \le C_0\lambda$. For

$$A_0 \ge \max(C_0, 2D), \qquad \lambda \le \frac{\theta}{2CA_0},$$

induction gives

$$\boxed{p_k \le A_0 \lambda \theta^k.}$$

Each term consumes at most half the next bound. The scale errors are summable when $1 < z < \alpha - d$, possible precisely for $\alpha > d + 1$.

6 Finite seed, growth and local delivery

A large initially perfect repetition code would evade the central question. Instead start from a fixed finite seed and, if needed, a fixed writer. Its exposure costs at most $C_{\mathrm{prep}}\lambda$ plus its declared entry error.

At level k, an ordinary local child circuit copies the current logical bit into five Info positions of a level-$(k+1)$ block. It includes reset, transport, waiting identities and guards. At most $M(B) = O(B^2)$ child spacetime locations suffice, a fixed family constant. Its failure is at most $M(B)p_k$ by a union bound.

Unused hardware may start arbitrarily. A constant reset puts zero into every clean Hold pass regardless of incoming Info, so a nice reset produces regular zero. Warm scale 1, then scale 2, and so on. Lower failures stay in the support envelope; one does not demand that every microscopic reset succeed. Parallel warmup costs $O(\sum_{j\le k}\tau_j) = O(\tau_k)$ while the retained source executes a fixed number of protected level-k identities.

The destination is fresh and disjoint from the source; reset masks never erase the source. Successive adjacent intervals occupy only a geometric sum $O(L_K)$. Deterministic phase padding aligns all child macro-boundaries, with preceding-period guards retained.

The precise adapter identities are

$$C^{k+1}(f) = C^k(C(f)), \quad \mathrm{up} = C^k(\mathrm{Enc}_B), \quad \mathrm{down} = C^k(\mathrm{Dec}_B).$$

They use $O(B)$ child depth and $O(B^2)$ child spacetime. The lower-level ports and decoders do not change across an adapter; only the outer grouping changes.

When its finitely many coarse locations work, an upward conversion produces the exact next-level projected encoding. Lower physical residues remain in the support field and guards. Downward conversion locally collects five child values, votes, and retains one child encoding, with the same error bound. Iteration reaches a fixed finite terminal circuit. No global decoder, external backup or early protected transcript is used.

Given T, choose K with $\tau_K \ge T$ and $\tau_K = O(T+1)$. Ascend, hold as needed, then descend. Delivery takes $O(T+1)$, including possible $O(T)$ readout delay; the bit remains exposed until delivery. For fixed starting level k_0, total error is at most

$$\delta_{\mathrm{prep}} + 2M(B)A_0\lambda\frac{\theta^{k_0}}{1-\theta} + C_{\mathrm{hold}}A_0\lambda\theta^K + \delta_{\mathrm{terminal}}.$$

Choose fixed writer/terminal budgets and sufficiently small positive intensity to make this at most $1/4$, uniformly in T. This completes Theorem 1.

6.1 The execution bill

Resource	Bound or premise
Input seed	Fixed finite radius independent of horizon
Active binary line	$O(T^{1/z}+1)$ mutable cells
Delivery	$O(T+1)$ including local decoding
Readout latency	Can be $O(T)$; no fast-query claim
Local work	$O(T^{1+1/z}+1)$ including waiting, reset and refresh
Schedule	Uniform compilation polynomial in produced circuit size; explicit list costs work times logarithmic addressing overhead
Energy	Sum of charged construction and finite per-primitive costs; no thermodynamic optimum
Protection	Every mutable bit corruptible; immutable laws and schedule granted
Reuse	Paid reset/write procedure for each input; no protected archive

The active-line bound is not a claim that an infallible physical controller fits inside those cells. Its description is charged; its reliability remains the approved mathematical premise.

7 P2: reset without hidden-force identification

P2's entropy composition and amortized waiting results remain valid. Its quadratic waiting is sufficient; its lower example restricts timing/retention, and its hidden-force information bound is not an intervention theorem.[1]

Consider its source

$$dx = p/M\,dt, \quad dp = (-kx + F - \gamma p + u)\,dt + \sqrt{2\gamma\sigma^2}\,dW, \quad \sigma^2 = M/\beta.$$

The inaccessible F is constant during preparation. Write $k = M\lambda_s(\gamma - \lambda_s)$, $0 < \lambda_s \le \gamma/2$, to distinguish the slow rate from the burst intensity. A finite position instrument and additive force actuator are explicit access premises. The final route needs no pin or curvature control.

7.1 Exact reference-preserving Gaussian replacement

Let U include initial source and frozen hidden variables, jointly distributed with arbitrary old reference R. The completed observation and prospective uncontrolled endpoint are

$$O = GU + \nu, \qquad (x_T^0, p_T^0)^T = HU + (N_x, N_p)^T,$$

with centered joint Gaussian noise independent of (U, R). It includes meter backaction and later bath noise. If h_p is the momentum row of H, choose $G^T w = h_p^T$, and set

$$Z_p = N_p - w^T\nu, \quad V = \operatorname{Var} Z_p, \quad c = \operatorname{Cov}(N_x, Z_p), \quad r = \sigma^2 - V > 0.$$

Draw fresh $D_p \sim N(0, r)$, set $D_x = -cD_p/r$, and use an additive force with endpoint displacement

$$(\Delta x, \Delta p) = (D_x, -w^T O + D_p).$$

Then $p_T = Z_p + D_p$ and $x_T = h_x U + N_x + D_x$. Momentum is thermal, has no initial-state gain, and has zero covariance with the final position noise. Joint Gaussianity proves

$$\boxed{\mathcal{L}(p_T, x_T, Z, R) = G_\sigma \otimes \mathcal{L}(x_T, Z, R).}$$

Initial laws can be singular or non-Gaussian. Second moments are needed for expected control cost, not the identity. The position marginal changes during preparation, and the target uses its contemporary value. Fine observations and random seeds must be discarded/isolated or reset; retaining them breaks the identity.

For $S = \operatorname{Cov}\nu > 0$ and $b = \operatorname{Cov}(\nu, N_p)$, the constrained minimum-variance predictor is

$$w = S^{-1}b + S^{-1}G(G^TS^{-1}G)^{-1}(h_p^T - G^TS^{-1}b).$$

This is established Gaussian estimation/covariance-control machinery.[5] The supplied bridge is the uniform old-reference identity with unknown force, physical finite sensing, backaction and causal endpoint control.

7.2 Finite meter, finite force and uniform feasibility

In units $M = \gamma = \sigma = 1$, use eight pointers with free Hamiltonian $P_j^2/(2m_{\mathrm{ptr}})$ and successive couplings $g_j(t)xP_j$. Thus $\dot{Q}_j = P_j/m_{\mathrm{ptr}} + g_j x$, $\dot{P}_j = 0$, and the source receives $-g_jP_j$. These effects are included.

The nominal meter takes time 1, with eight boxes of width 1/8, gain 1, pointer mass 100, $\operatorname{Var}Q_j = 10^{-6}$ and $\operatorname{Var}P_j = 10^{-2}$. Read at the common deadline and apply force for $h = .2$. Exact matrix propagation and 100 directed-interval covering pieces prove

$$\sup_{0\le k\le 1/4} V_{\min}(k) < .767447 < 1.$$

Point optima are about .619–.624. The certificate uses exactly mean-correct candidate weights, not floating minimizers as proof. $k = 0$ is a limiting certificate; every original physical member has $k > 0$.

Two disjoint $\sin^2$ pulses, one on each half of the final interval, have zero value and derivative at their endpoints. Their endpoint columns are

$$L_i = \int_0^h e^{A(h-s)}B_0 g_i(s)\,ds, \quad A = \begin{pmatrix} 0 & 1 \\ -k & -1 \end{pmatrix}, \quad B_0 = (0,1)^T.$$

Uniformly, $\det L \ge .795^2h^3/32 > 0$. Coefficients $L^{-1}(\Delta x, \Delta p)^T$ realize the displacement with no delta kick. Nominal reset time is $1.2/\gamma$. At $D_0 = 1$, expected normalized $\int u^2dt$ is about 224–233, or Girsanov action 56–58, excluding meter/cooling/records.

For P2's hidden sectors, the full augmented path still obeys

$$D(P_+^t \| P_-^t) = \frac{4D_0\lambda_s}{\gamma} + 2D_0\lambda_s(1 - \lambda_s/\gamma)^2 t.$$

Common adapted controls cancel from the drift difference. At fixed reset time, this information tends to zero with λ_s while replacement stays exact. The controller removes dependence without identifying the hidden sign.

7.3 Direct preparation of the useful sine arms

The stronger endpoint avoids a second marker. Put

$$q = \operatorname{Cov}(\nu, Z_p) = b - Sw, \qquad c = \operatorname{Cov}(N_x, Z_p).$$

Choose s with $s^Tq = -c$. Then $N_x + s^T\nu$ is independent of Z_p. Impose also $s^TG_x = 0$, where G_x is the initial-position column; this avoids a feedback displacement proportional to the diverging initial position. The first and last observations suffice to solve these two constraints.

Choose a fresh fair arm $A \in \{-1, +1\}$, withheld until the endpoint. With $r = \sigma^2 - V$, draw independent dither of density

$$f_A(d) = G_r(d)\,[1 + Aa \sin(\sigma d/r)]\,, \qquad a = \eta e^{V/(2r)} \le 1,$$

where G_r denotes the centered Gaussian with variance r. Apply the finite force giving

$$(\Delta x, \Delta p) = (s^T O, -w^T O + D).$$

Now $x_T = (h_x + s^T G)U + N_x + s^T \nu$ and $p_T = Z_p + D$. Gaussian convolution proves

$$\mathcal{L}(p_T \mid A) = G_{\sigma^2}(p)[1 + A\eta \sin(p/\sigma)] = q_A(p),$$

and, crucially,

$$\boxed{\mathcal{L}(A, p_T, x_T, Z, R) = \tfrac{1}{2} q_A(p_T)\mathcal{L}(x_T, Z, R).}$$

This is the original sine-family intervention at a fixed nonvanishing strength, not merely a mean-response match. It holds for arbitrary initial histories within the affine constant-force class. The hidden marginal is common to both arms and is the contemporary prepared marginal.

For the final implementation replace each rectangular pointer coupling by a C^1 pulse. In its $1/8$ window, it rises over $1/64$, stays at one for $3/32$, and falls over $1/64$, using $g(v) = 3v^2 - 2v^3$ on the rising edge and its complement on the falling edge. The normalized peak is one and maximum slew is 96γ. Keep the force window $.1/\gamma$ and choose $\eta = 1/2$. A 100-piece directed-interval certificate gives

$$V/\sigma^2 \le .542831311, \qquad a \le .905327083 < 1,$$

and the determinant of the two-coordinate position correction stays below $-.118702839$. Exact lifted matrix exponentials handle the cubic ramps and thermostat covariance, without time discretization. The point fixtures have $V/\sigma^2 \simeq .427$–$.429$ and $a \simeq .728$. The nominal operation takes $1.1/\gamma$; it does not claim every prescribed $\eta < 1$ with these particular hardware constants. Bounded slew does not mean strict Fourier band limitation.

Rejection sampling from G_r realizes f_A with acceptance probability $1/(1 + a)$, independent of the arm and source. Finite precomputed attempts, fixed deadlines and explicit failure branches make sampling cost visible. Fine records and seeds are isolated/reset before the endpoint is reused.

The faster construction is expensive: at $D_0 = 1$, point-fixture expected normalized $\int u^2 dt$ is about 13,324–23,324. These are not joules, mechanical work or an optimum. They describe the constrained-optimal point fixtures; the interval proof uses its own mean-correct candidate predictor, whose action is finite and bounded on the compact stiffness range, but no calibrated numerical uniform action bound is asserted. The position displacement has exactly zero initial-position gain, preventing a hidden $1/k$ divergence. A longer optimized controller or Gaussian reset plus marking can cost much less.

7.4 Response time and the full physical bill

After the direct intervention, the bare harmonic response has position-increment contrast

$$\Delta\mathbb{E}[x(t) - x(0)] = 2\eta\sigma e^{-1/2} \frac{e^{-\lambda_s t} - e^{-(\gamma - \lambda_s)t}}{M(\gamma - 2\lambda_s)}.$$

At $t = 1/\gamma$ the gain tends to $(1 - e^{-1})/(M\gamma)$ as $\lambda_s \to 0$ and is $e^{-1/2}/(M\gamma)$ at critical damping. Useful response occurs on the fast friction time. A preparation wait $1/\lambda_s$ would dominate this experiment; the direct controller avoids it. Actual finite readout/backaction and statistical repetitions remain charged.

For a calibrated finite marker $Y = p + \xi$, independent Gaussian error of variance v can also be compensated exactly by

$$a_v(Y) = \tfrac{1}{2}[1 + \eta e^{v/(2\sigma^2)} \sin(Y/\sigma)]$$

when it lies in $[0, 1]$, provided the error is independent of the full physical endpoint noise as well as the trigger state. Thus fixed MSE alone is not a lower bound on accumulated marking loss. The direct-arm route above avoids needing that second sensor.

For an imperfect Gaussian reset endpoint with momentum mean $B\theta$ and noise covariance $\left(\begin{smallmatrix} V_x & c \\ c & V_p \end{smallmatrix}\right)$, the augmented replacement divergence is

$$J = \tfrac{1}{2}\left[\frac{\mathbb{E}(B\theta)^2}{\sigma^2} + \frac{V_p}{\sigma^2} - 1 - \log\frac{V_p - c^2/V_x}{\sigma^2}\right].$$

Near the exact endpoint it is quadratic under bounded moments and covariance margins. An $O(\epsilon/\sqrt{n})$ coefficient error can suffice for campaign KL $O(\epsilon^2)$, but acquiring and maintaining that accuracy requires an actual calibration protocol.

The nominal $1.1/\gamma$ assumes that the completed pointer observation is available when the force window starts. Readout and computation latency must be accommodated causally. One option reserves an idle subinterval inside the final $.1/\gamma$ and shortens the two force pulses, preserving the source-noise prediction horizon at greater control cost. Continuing pointer drift or extra readout noise changes the observation matrices and requires a new certificate. Appending latency after a fixed endpoint calculation would be invalid. No actual ADC latency or precision specification is supplied.

A faithful calendar includes

$$T_{\text{cal}} + \sum_j [T_{\text{locate/center},j} + T_{\text{pointer prep/reset},j} + T_{\text{causal meter+force},j} + T_{\text{response/read},j} + T_{\text{disposal/logic},j}],$$

plus deadline, overflow, retry and record-protection costs.

The source position range grows as $\lambda_s^{-1/2}$. Differential centering can reduce signal range but requires a real sensor, positioning and memory. The pointer fixture requires a cold or nonequilibrium preparation; a thermal realization has temperature 10^{-4} of the source temperature, mass $100M$ and stiffness $100M\gamma^2$. Its phase-space action must remain in the classical regime relative to $\hbar$. Cooling and reuse are not free.

Gaussian commands have unbounded support: finite peak range and slew require tail allowances, with different bounds for second-moment versus sub-Gaussian inputs. Finite ADCs, arithmetic, random-bit generation and waveform synthesis are also priced. Discrete seeds are compared through the noisy force channel, not by asserting small TV to a continuous Gaussian.

The rectangular base-reset certificate is retained for comparison. The stronger direct-arm construction now has explicit C^1 meter and force couplings. An independent full 19-state calculation agrees with the smooth-meter covariance formulas to below 2.6×10^{-11} at a checked stiffness; the interval proof supplies the uniform statement. No actual calibration, recycled-pointer demonstration or laboratory trial was performed.

The endpoint identity composes with arbitrary later adaptive kernels while each next preparation returns to the stated affine/frozen-force contract. It is not the universal nonlinear campaign theorem. It refutes a force-learning lower-bound route, while leaving optimal total resources open. For a linear mean-response task, ordinary private force probes are an admitted cheaper comparator.

8 P1: actual matrices obstruct two sampler guarantees

The preceding full-syndrome history compiler, positive weighted extension and unknown-reference loading identity remain accepted. The new comparison uses the actual BDL RX/RZ family at fixed odd subdivision parameter $\ell \geq 3$, not a surrogate parity-history toy.[7]

Let $M_Z(L)$ be the binary matroid represented by physical Z-check columns of $H_Z(L)^T$. Its dual is generated by the physical dependency space $\ker H_Z(L)^T$.

Theorem 2 (Extensive Fano obstruction)**.** *For even $L \geq 6$, the dual Z-check matroid has a minor $\bigoplus_{i=1}^{r_L} F_7$, where*

$$r_L \geq \frac{N_L}{1680 C_\ell^4}, \qquad C_\ell = 100((\ell-1)/2)^2.$$

The primal has the dual direct-sum minor. Both graphic and cographic defects in the Code Swendsen–Wang criterion are at least r_L, and the physical Z-check matroid is nonregular.

8.1 The physical-column certificate

An RX dependency has independent parameters: inherited d, one free s_z per old Z check, an even vector a_q on each qubit's Z neighbors, and a cycle F_x on each X-link graph. Actual physical check columns carry

$$s_z,\ s_z + d_z; \qquad a_{qz},\ a_{qz} + d_z,\ a_{qz} + s_z; \qquad F_{xf},\ F_{xf} + d_z,\ F_{xf} + a_{qz}.$$

Interior subdivision equations repeat these coefficients; the table exhausts the physical nullspace. The report gives the cell-by-cell proof and exact checks.

A true RX corner, preserved through RZ, supplies the wheel with edges

$$AB,\ AC,\ BC,\ AD,\ BE,\ CD,\ CE,\ DE.$$

Explicit physical-column contractions leave inherited values

$$(0, w, 0, w, 0, w + u, u, u),$$

where $u = s_z$ and w is a local even-star mode. The third RX supplies an independent wheel cycle. Contracting $u + w, F'_{AC}, F'_{BC}$ leaves seven actual columns

$$w,\ x,\ x + w,\ x + t,\ x + t + w,\ t,\ t + w,$$

all nonzero vectors in $\mathbb{F}_2^3$: an F_7 minor.

Terminal, branch and face columns span all eliminated parameter spaces, so the operations are genuine matroid contractions. Vertex-disjoint qualifying squares give independent local (u, w, F') spaces. Bounded Tanner degrees supply at least $N_j/1680$ such squares. Four bounded replacements compare N_j with N_L, yielding the extensive bound for every fixed allowed ℓ.

If a matroid contains r independent nonregular minor blocks, a projection to a regular matroid must lose at least r ranks: its kernel must intersect each block span nontrivially, giving independent lost directions. A Δ-graphic certificate is precisely a projection to a graphic representation losing at most Δ ranks; apply this also to the dual.[8]

Therefore neither orientation has the logarithmic defect required by the published polynomial CSW guarantee, and the regular-matroid Ising FPRAS lacks regularity.[9] This excludes those sufficient routes, not CSW itself, all efficient samplers or all tensor algorithms. Both defects refer to the Z matrix and its dual, not an unproved statement about both CSS sectors. One obstructed sector suffices to prevent those direct joint-Gibbs guarantees.

The coefficient is tiny at BDL's enormous fixed embedding parameter. No practical runtime separation, hardness theorem, general coherent robustness or actual passive quantum device is claimed.

9 Verification and retained boundaries

The package contains complete construction notes, formal interface proofs, independent audit, exact finite checks, interval certificates, original governing returns and this report's source. It distinguishes proofs, numerical receipts and hardware premises.

Check	What it establishes
P3 semantic check	491,520 adversarial active-pass cases and 640 boundary cases for the two-round identity core; generic fixed-fanin correctness is proved separately
P3 independent audit	Synchronized commits, arbitrary-input normalization, raw influence boxes, Poisson separation, staged prewarming and scale interfaces
P2 Gaussian checks	Independent full/reduced pointer matrices, causal endpoint identities and finite-pulse synthesis
P2 intervals	Uniform feasibility for reset and fixed-strength direct sine arms on $0 \le k/(M\gamma^2) \le 1/4$
P1 exact binary checks	Literal RX/RZ incidence, full dependency equality, three independent wheels and rank-nine Fano direct sum
P1 subdivision checks	Actual cellular constructions at $\ell = 3, 5, 7$; the all-ℓ conclusion follows from the proof

There were zero physical trials. A finite checker does not prove an infinite hierarchy; the hierarchy rests on the deterministic compiler and marked-Poisson argument. Floating covariance checks do not replace the interval inequalities. The report does not establish historical priority: it explicitly credits Gacs, Tsirelson, extended-rectangle composition, Gaussian output-feedback control, Code Swendsen–Wang and regular-matroid sampling.

The strict-Pareto target is resolved within the declared model. The general finite-$(d+1)$-moment endpoint, an autonomous mutable controller, a passive quantum analogue, a practical apparatus, and a universal nonlinear intervention-cost optimum remain separate questions.

The prior equilibrium-inference and quantum documents are preserved, not silently strengthened or replaced. The contribution to discovery capability is a reusable proof interface: semantic correction maps arbitrary damage into a local geometric fault field; geometric noise can then be analyzed without assuming independent error values or an ideal final decoder.

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